

Cross-Chain Trustless Credit Bureau & 5-Track Attestcoin Engine

Built natively for the Creditcoin BUIDL Hackathon 2026 Fall (DoraHacks)
Deeply integrated with Creditcoin's Attestcoin Protocol (precompile 0x0FD2) / USC

LIVE 0x0FD2 integration | Run live: credx-protocol.vercel.app | 126/126 tests | 22 contracts on testnet

Why DeFi treats proven borrowers as strangers

The bottleneck

- DeFi lending demands 150%+ over-collateralization - to borrow \$1,000 you must lock up \$1,500.
- Borrowers who repaid hundreds of thousands on Ethereum (Aave, Compound, Uniswap LP) or settled trade invoices start from zero on every new chain.
- Multi-sig bridge relayers have suffered over \$2.8B in exploits across Web3.

CredX answers

- Creditcoin becomes the global trustless credit bureau - reusing verified cross-chain history.
- Lend down to 70% collateral ratio (53.3% less capital locked).
- Dynamic FICO-style APRs from 2.5% to 12% based on real on-chain reputation.

Zero bridges, zero custodial relayers, zero centralized oracles

Direct precompile verification

- BlockProverAttestationOracle wraps Creditcoin's BlockProver 0x0FD2 + ChainInfo 0x0FD3 precompiles
- using the exact @gluwa/usc-sdk ABI (MerkleProof + ContinuityProof structs).
- Verifies real Merkle Patricia Trie inclusion proofs from Ethereum Mainnet and Sepolia.
- verifySourceTransaction (view, static) and anchorVerifiedTransaction (verifyAndEmit + anchor).

LIVE proof transcript (Creditcoin testnet)

```
Verified SUCCESS on 0x0FD2 - real Sepolia tx 0x1eb62c...72d7af
verifyAndEmit (TransactionVerified) 0x7dff1e...b0dd29
Oracle anchor (anchoredCount=1)    0xd0b88f...befd69
```

Reproduce any time: npm run usc:verify

Proposals proposed; the deterministic policy decides

- Four-gate trace before anything executes: (r1) collateral liveness, (r2) covenant boundary,
- (r3) policy action allowlist (fail closed), (r4) 0x0FD2 oracle-bound cryptographic verification.
- Failing gate => explicit REFUSED with the violated rule cited; no silent partial execution.
- Live in-browser 0x0FD2 button verifies the latest attested Sepolia receipt against the
- deployed precompile - Merkle + continuity proof, no wallet, no signer, no gas.

Every verified spine fact in one pane of glass

- Attestation heights, anchored proof count and source chains read live from the deployed oracle
- (0x0FD2 + 0x0FD3 ChainInfo) - no operator, no bridge, no oracle.
- Collateral liveness rules decide whether new credit stays open per position.
- On attested collateral departure: new credit is blocked; repay/withdraw stay open;
- a fresh receipt proving restoration re-opens credit automatically.

Money locked to a declared purpose until attestation unlocks it

- PurposeBoundFunding: funds are bound to a purpose code at funding time (invoice purchase,
- equipment, payroll, GPU lease...) - the purpose cannot be changed after funding.
- Until an attested usage receipt is verified, disbursement goes ONLY to the allowlisted
- counterparty - never to the borrower; attested tranches then unlock.
- Covenant deadswitch: a proven breach freezes the record and blocks all further disbursement;
- settlement repays principal + interest and refunds collateral. Deployed & verified on testnet.

Usage is attested, capped, priced and prepaid-first on-chain

- UsageMeteringRegistry (prepaid v2): meters per wallet + action key (GPU lease, flash-loan cycles, compute ops, arena ticks) configured by the registry owner with caps and unit prices.
- Meter advances only on a verified attested receipt (public or KYC-agent telemetry path).
- Window caps fail closed: EXCEEDS WINDOW CAP reverts the increment; debt accrues on-chain and is settlable in cUSD.
- v2 prepaid credits: a settlement-token top-up funds a prepaid balance consumed BEFORE debt;
- while prepaid exists, overdraw reverts INSUFFICIENT PREPAID (fail-closed). Deployed & verified.

Proof-gated settlement and a dispute-ready audit surface

- VerifiedEscrow: depositor locks cUSD against an order; release ONLY on a cross-chain receipt
- verified against the Attestcoin verifier (optionally pinned to the exact event signature).
- Seller / order ref / amount / deadline bound at creation; keccak(sourceChainId, txHash) nullifier
- makes each receipt spendable ONCE across escrows (replay-guarded).
- After the deadline anyone triggers the refund - funds can never be locked forever.
- Evidence Registry: ProofAnchored, EscrowReleased, UsageRecorded and PrepaidConsumed events
- recovered live via eth_getLogs from the deployed contracts - the exact read surface an auditor uses.

Academic On-Chain Credit Risk framework

- 7-dimension model: volume, frequency, recency, chain diversity, protocol diversity, action weights.
- Composite Credit Trust Score (CTS) from 300 to 850 per wallet.
- Deterministic keccak256(chainKey|height|encodedTx) replay protection - no double credit ever.
- CX-SBT (ERC-721 soulbound): non-transferable tier passport with ZK-commitment privacy.

CredX 5-Track Attestcoin Super-Ecosystem

- Under-Collateralized Lending Pool: Super-Prime (CTS 780+) borrow cUSD against CTC at 70% collateral (vs 150%).
- Reputation Flash Loans (ReputationFlashLoan): uncollateralized loans, Super-Prime fee 0.01% (vs 0.09%).
- Reputation AMM fee tiers down to 0.05% (from 0.30%); Yield Vault up to 2x reward multiplier.

CredX 5-Track Attestcoin Super-Ecosystem

- RWA Invoice Financing (RWAInvoiceFinancing): tokenize trade invoices; prime investors fund at preferred rates.
- Treasury Yield Fund: permissioned tbUSD vault with +2% APY bonus for prime depositors.

CredX 5-Track Attestcoin Super-Ecosystem

- Anti-Sybil Proof-of-Humanity: up to 3x daily gathers tied to verified cross-chain credit.
- Sybil-proof lootboxes: high-tier opens require CTS ≥ 500 .
- Zero-collateral NFT guild scholarships (0% upfront) + zero-fee marketplace for Super-Prime.

CredX 5-Track Attestcoin Super-Ecosystem

- Automated staking delegation only to operators with CTS ≥ 700 (slashing protection).
- Under-collateralized hardware financing for verified operators.
- Chrome Extension Virtual Node (Manifest V3): local telemetry + MetaMask relay for real proof anchors - never stores keys.

CredX 5-Track Attestcoin Super-Ecosystem

- Oracle-less risk ingestion: AI models read multi-chain proofs to tune protocol parameters.
- AgentFi credit lines for registered on-chain AI agents with verified execution records.
- Verifiable Proof-of-Compute escrow: GPU leases settle on cryptographic proof verification.

ReputationArena - binary paper trading terminal

- \$10,000 paper trading points, 60-second round intervals, on-chain settlement.
- Achieve a 3+ win streak and sync it to Creditcoin L1 for a permanent on-chain credit boost.

126/126 automated tests

- npx hardhat test -> 126 passing (BlockProver oracle 9, hub/engine 26, security 19, tracks 45,
- policy layer + prepaid 19, verified escrow 8).
- OCCR math, dynamic APRs, CX-SBT soulbound invariants, replay defense: all covered.

Clear labeling of simulations

- Panels that are locally simulated (perps, flash-loan UI, liquid staking, TAO/GeoOrbit telemetry,
- liquidation radar, DePIN telemetry, attestation pipelines) carry explicit SIMULATED badges.
- The real 0x0FD2 verification path is live in the 'Proofs & Attest' tab; the Mock harness is
- a clearly labeled gasless fallback for score boosts only.

22 deployed & verified contracts

BlockProverAttestationOracle (real 0x0FD2/0x0FD3)

0x4d11b60809724b0B67B28DA2f38438aE97f1C671

AttestationVerifier (labeled testnet harness)

0x34aA30efE2226ffC2E55607017FbA2F07e62b279

CreditScoreEngine (OCCR 7-Dimension)

0xA31697b8d4900f8FA62015A51dA3c58972E96BB6

CredXHub (Core Registry & Batch Importer)

0x729b2D88630c4241d051c92D4Fe831412846eE18

cUSD (Liquidity Stablecoin)

0xddec5170C46DC63D812c699E9dFE6561FFd1BF298

UndercollateralizedLendingPool

0x84234C1403768D9A509c1241e5F39f81246880d2

CreditAttestationSBT (CX-SBT)

0xb22baF385067aF8d66823282bb4F2e3EECB60831

ReputationAMM / FlashLoan / YieldVault

0x8146...32c / 0x4962...637 / 0x6309...bc1

RWATreasuryYieldFund / RWAInvoiceFinancing

0x2be1...80A / 0x05D4...1aA

GamingEcosystemHub / DePINInfrastructureHub

0x8008...4D75 / 0x99b4...B98c

AutonomousAIHub / ReputationArena

0xEc14...a5D / 0x42ff...c9F5

PurposeBoundFunding (purpose-bound RWA vault)

0x551592C32a96555A04BB016c2DF7138A1f9DE644

UsageMeteringRegistry (prepaid v2: metered usage)

0xF8a9645ac3D234cf72B0C4C170cFB289FE2Ae4F9

VerifiedEscrow (condition-locked, proof-gated settlement)

0x07aBcbb7b2F9f4400c93d092F343e186ee526137

React 18 + TypeScript + Vite + Three.js

- Web3 Terminal: RWA invoices, DePIN telemetry, flash loans, liquidity depth, reputation arena.
- Proofs & Attest tab: live 0x0FD3 chain reads + connected-wallet 'Verify & Anchor on 0x0FD2' button
- that verifies a fresh Sepolia proof in-browser (official proof-builder API + ethers, no SDK bundle).
- Policy panes live on-chain: Purpose-Bound RWA, Usage Meters + Prepaid, Verified Escrow (real
- release/refund/top-up transactions with a connected wallet) and the Evidence Registry (eth_getLogs).
- Chrome Extension Virtual Node: load unpacked from chrome://extensions.

Everything you need to verify

Commands

<code>npm install && npx hardhat test</code>	<code># 126/126 tests</code>
<code>npm run usc:verify</code>	<code># live 0x0FD2 Sepolia proof + anchor</code>
<code>npm run usc:deploy</code>	<code># deploy BlockProverAttestationOracle</code>
<code>npm run deploy:policy / deploy:flagship</code>	<code># policy layer + escrow on testnet</code>
<code>cd frontend && npm install && npm run dev</code>	<code># Web3 Terminal (localhost:5173)</code>

Links & license

- GitHub: <https://github.com/tondays52/CredX>
- Live app: <https://credx-protocol.vercel.app> (RiskGuard, Covenant Ops, Purpose-Bound RWA, Usage Meters +
- Prepaid, Verified Escrow and Evidence Registry panes live on-chain)
- License: MIT (open source for the Creditcoin L1 ecosystem).